Program Update Day April 10 2019		**UU Supervisor**
Location: BOL - 1.022: https://students.uu.nl/bolognalaan-101		
13.15- 13.30	*Welcome*	
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13.30 - 13.45:	Annemieke	David
13.45 - 14.00:	Willem de B	Joke
14.00 - 14.15:	Willem vd M	Peter
14.15 - 14.30	*Break*	
14.30 - 14.45:	Bas	Chris
14.45 - 15.00:	Sjors	Sita
15.00 - 15.15	Niilo	Anouk
15.15 -16.15	Posterpitches + questions at the posters -> max 5 min per poster	
16.30 - 16.45	Laura	Nico
16.45 - 17.00:	Mitchel	Marnix
17.00 - 17.15	Jasmine	Krista
17.15 - 17.30	Frits	Stefan
17.30 -18.30:	*Drinks*	

Posters	**UU Supervisor**
Alex	Estrella
Elles (?)	
Floor	Stella
Kazuya	Anouk
Margreet	Ignace
Mathilde	Nico
Roderick	Chris
Stefan	Anouk
Thomas	Martijn

PPT Presentations

Name Student: Annemieke San Giorgi
Title project: Search system for the Inclusie Toolkit
Module: Thesis
Abstract: Recently a new law stated that all governmental services should be accessible to the entire population of the Netherlands, including people that have problems with digital processes, such as elderly, illiterates and blind people. A next step on this front is inclusive design, which not only ensures accessibility, but aims for a pleasurable experience for everyone. Since information on inclusive design is scarce, the project Dutch Inclusive Design for IT (DID-IT) of TNO Soesterberg aims to create a toolkit that can help all stakeholders of inclusive design projects. Information in the toolkit will be available on raising awareness on the importance of inclusive design, the application of inclusive desing and sharing experiences with inclusive design. Since there will be large quantities of information available in this toolkit, some sort of search system needs to be implemented, in order to have a high level of usability for its users. The aim of this thesis has been to design such a search system that can fulfil the needs of the wide variety of users that will use this toolkit and will all need different types of information. Through interviews and ther types of analyses, the design requirements have been put together. A desgn for a search system has been protoyped, consisting of a filtering system for specific search queries and a categoric search system that supports explorative search. Through research, the optimal filters have been picked for in this system, since this is an important factor for success. After that, a usability test has been performed to validate the efficiency and usability of this search system.
Location project: TNO Soesterberg
External supervisor: Rosie Paulissen
UU Supervisor: David Terberg

Name Student: Willem van der Maden
Title project: Green relaxes, red arouses? An essay on the challenges of color emotion research
Module: Internship
Abstract: Color and emotion are two concepts that are a part of our everyday life. Research has shown that there is a strong relationship between both concepts that resulted in a research field called color emotion. Color emotion essentially refers to the experience of a certain emotion after or whilst viewing a specific color. From investigating the literature, it became apparent that there are two separate scientific discourses that research color emotion with different methods. The first discourse is centered around an article by Valdez & Mehrabian (1994) and the second discourse is centered around an article by Sato, Kajiwara, Xin, Hansuebsai, & Nobbs (2000). The aim of this essay is to investigate what challenges in color emotion research exist in the current paradigm, and how they have been

managed. I have identified the starting point of the current paradigm to be an article by Gelineau (1981), where she identified several problems with color research. Ensuing these findings, Valdez & Mehrabian (1994) separate the problems into two categories: methodological issues and conceptual issues (Valdez & Mehrabian, 1994). The issues proposed are mainly with regard to the standardization of the stimuli and standardization of the method for recording emotional response. In this essay, articles from both previously mentioned discourses will be reviewed on how they manage the methodological and conceptual issues. In order to review the articles, two theoretical frameworks are provided. One framework regarding the terminology of color perception, the other framework regarding emotion. The reason for using a framework for color perception is to avoid confusion concerning the terminology of color perception. A framework for emotion is provided in order to provide background information on emotion. Explanation of the theoretical framework of color perception and emotion will be provided in the next section. After analyzing the articles, methodological issues that pose challenges to the research field can be identified. The first claim of this essay will be that the challenges caused by methodological issues can be resolved. The second claim will be that considering the appraisal theories of emotion and the current situation, color emotion research will not yield consistent results. The current situation refers to the useful findings that have been generated by the current literature and how the present challenges that arise from color emotion have been managed.
Location project: Omgevingspsycholoog Den Haag
External supervisor: Wouter Tooren
UU Supervisor: Peter Bos

Name Student: Willem de Bruijne
Module: Thesis
Title project: Suppression of theta oscillations in a stop signal task with informed differential probabilities
Abstract: The inferior frontal gyrus (IFG) has shown to be a key area for motor response inhibition. However, the underlying neural mechanism that results in appropriate motor inhibition is not completely understood. This study therefore aimed to further elucidate the necessity of theta-frequency oscillations on motor response inhibition, by attempting to suppress the theta activity using repetitive transcranial magnetic stimulation (rTMS) at the IFG. 28 volunteers performed a stop signal task during two separate sessions, of which one session contained the active, IFG rTMS protocol. The stop signal reaction time (SSRT) was calculated, which aims to quantify the time onset of response inhibition. Moreover, the effects of rTMS on the theta power (4-7.5 Hz) in the time-frequency domain of the EEG, and its relationship with the SSRT were assessed.
Location project: Utrecht University
External supervisor: -
UU Supervisor: Joke Baas

Name Student: Bas Oppenheim
Module: Thesis
Title project: Changing gears: Shifting attentional focus in a semi-autonomous vehicle.
Abstract: Semi-autonomous vehicles – i.e., SAE Level 3 – are becoming more and more available. Such vehicles are able to fully take over all driving tasks from the human driver, but will in specific circumstances, e.g. due to legal or technical limitations, signal the human driver to intervene, which they are then obliged to respond to. An issue specific to driving such autonomous vehicles is that it typically invites the human driver to distract themselves with non-driving tasks. Thus, when the vehicle signals the driver to intervene, they should ideally be given ample time to 1) break away from their non-driving task and 2) reorient themselves to the traffic situation, both of which help increase attention to driving. To better understand attentional processes around a semi-autonomous vehicle's signal to intervene, Janssen & Donker (draft) suggest a framework in which the intervention event actually consists of 10 distinct sub-events, or 'stages of transition', some of which overlap. Because there is no quantified data available, in this study we set out to find information about these stages, including driving and non-driving task performance, stage duration and how they may influence each other. We used a semi-autonomous car simulator, equipped with all hardware typically available in the driver's seat, to recreate the intervention signal scenario. We invited people with driving experience to participate as the driver. The intervention signal typically came after around 3 minutes of autonomous driving, signalling the driver to take over control in order to swerve around two stationary cars. After evading the cars, the driver could return control to the car. Meanwhile, to measure non-driving task performance, the driver was tasked with filling in a calendar, with calendar events consisting of either 2 (quick to finish) or 6 (takes a while to finish) items. Using this scenario simulation, driving performance was scored by safety and steering movements, while non-driving performance was scored by the number of completed calendar items and errors. In addition, the point of breaking away from the calendar task was also analysed, as was the time it took to respond to the signal. The data suggest that breakpoints play a major role in determining the duration to take over control. Possible implications for safe driving behaviour in and of semi-autonomous vehicles are discussed.
Location project: UU
External supervisor: -
UU Supervisor: Chris Janssen

Name Student: Sjors van de Ven
Module: Thesis
Title project: Why don't they sleep? Spectral analysis of the initial non-rapid eye movement (NREM) sleep cycle in patients suffering from insomnia.
Abstract: Background: Although the etiology of insomnia has been investigated substantially, no consensus has been found regarding the characteristics of cerebral activity during sleep of insomnia patients Objectives: The objective of the study is to contribute to

the elaboration and clarification of the physiological understanding of pathological sleep in insomnia. In order to do so, the study will focus on the beta frequency activity, a biomarker for cortical activity and arousal. Subjects: 52 insomnia patients and 48 age- and gender-matched control participants were measured two consecutive nights in order to exclude the 'first night effect'. Methods: Small windows of sleep (epochs) were scored according to the clinical guidelines of American Academy of Sleep Medicine. The power spectral analysis was calculated from the continuous sleep-EEG data by using Fast Fourier Transformation. Design: Sleep characteristics during the first non-rapid eye movement sleep cycle of insomnia patients and a control group, as assessed by spectral power, were compared between groups. Statistics: An independent samples t-test was used to investigate the differences in sleep characteristics between insomnia patients and controls. Expected results: The insomnia patients were expected to show an enhancement in beta frequency spectral power compared to the control group, a finding that would support the 24-hour hyperarousal theory as a major factor contributing to the etiology of insomnia.
Location project: Netherlands Institute for Neuroscience
External supervisor: Jennifer Ramautar
UU Supervisor: Sita ter Haar

Name Student: Niilo Valtakari
Module: Thesis
Title project: An eye tracking approach to Autonomous Sensory Meridian Response (ASMR): The physiology and nature of tingles as seen through the pupil
Abstract: Autonomous sensory meridian response (ASMR) is a sensory phenomenon commonly characterized by pleasant tingling sensations originating from the crown of the head and accompanied by feelings of relaxation and calmness. Previous research has found ASMR to have a distinct physiological pattern with increased skin conductance levels and reduced heart rate. ASMR has also been shown to increase positive emotion while decreasing negative emotion, indicating that it could have therapeutic benefits. The aim of the present study was to investigate the physiology and characteristics of ASMR further, by examining whether experiencing ASMR is visible from the pupil of the eye. A total of 91 participants were recruited and assigned to three different groups based on their experience of ASMR (ASMR vs. non-ASMR vs. unsure). Participants were tasked to watch a control video and an ASMR video and instructed to report any tingling sensations by pressing down a button on the keyboard. Pupil size was measured over the duration of both videos using a mounted eye tracker, and analyzed on a general level, averaging pupil size over each video, as well as on a more specific level, comparing pupil size during reported tingling sensations to pupil size outside of those episodes. On the general level, results revealed no significant differences between the groups. On the specific level, however, the tingling sensations experienced in ASMR were found to cause statistically significant increases in pupil size, demonstrating that the tingling sensations experienced in ASMR have

a distinct physiological basis. The results of the study further reinforce the credibility of ASMR and suggest that the tingles in ASMR are at the very core of the experience itself.
Location project: Utrecht University
External supervisor: N/A
UU Supervisor: Anouk Keizer

Name student: Laura Osinga
Module: Thesis
Title project: The impact of Electronic Medical Records on the quality of communication between doctors and patients in the consultation room
Abstract: Background: Electronic Medical Records (EMR) are increasingly used in health care organizations and the implementation has changed the dynamics of doctor-patient communication. This transition requires research, since patient-doctor communication might be the most significant component of the healthcare visit, with ramifications for patient satisfaction, adherence to treatment, conflict resolution, laboratory costs and clinical outcomes (Duke et al., 2013, p. 358).
Objective: This article offers insights into the relationship between the amount of eye-contact between physician and patient and the extent to which the physician actively reacts to the cues and concerns of the patient during consultations.
Methods: 24 videotaped consultations of 6 different physicians were used to observe the cues and concerns of patients and the corresponding responses of the physicians. In addition, the eye-movements of physicians were measured with an eye tracker to scale the amount of eye-contact.
Results and conclusion: not yet finished.
Location project: AMC
External supervisor: MSc. Chiara Jongerius
UU Supervisor: Dr. Nico Romeijn

Name Student: Mitchel Kappen
Module: Thesis
Title project: Human Selection in a Job Recruitment Setting: A Video-Based Physiological Marker Model for Predicting Person-Organization and Person-Job Fit
Abstract: In human job recruitment, decisions on whether to hire someone are often based on subjective factors such as likeability or gut feeling. This has lead to differences in pay and unequal chances for people of different ethnicity, gender, and even physical attractiveness. We had students participate in an online assessment for a job, whilst their faces were recorded with a webcam. The video footage was analyzed and with the use of different algorithms, emotions, blinking patterns, and heart rate measures were extracted. These variables were used to develop a model that can predict crucial factors in employee quality and employee retention.
Location project: Neurolytics BV

External supervisor: Felix Hermsen
UU Supervisor: Marnix Naber

Name Student: Jasmin Gerritsma
Module: Thesis
Title project: Do you understand what I see? Which of four simulations of visual field is best at clarifying how people with a visual field defect experience the world?
Abstract: The goal of this study was to see what simulation of visual field defect is most effective at conveying the message it should convey, namely giving close friends and family members of someone with a visual field defect an idea of how a person with a visual field defect sees the world. The four simulations where taped of safety glasses, an augmented reality app on a smartphone, a small gaze contingent simulation on a 17-inch monitor and a large gaze contingent simulation on a projection screen. The effectivity was evaluated by means of a questionnaire. The large gaze contingent simulation has been found to increase understanding of the consequences visual field defects have on daily life. No other differences were found in the quantitative analysis. When asked what simulation they would use themselves to explain what their close friends/family member sees, most participants said they preferred the taped off safety glasses, followed by the simulation on the large gaze contingent simulation. The recommendation for Royal Dutch Visio is to keep using the modified safety glasses and to start using the large gaze contingent simulation. It is recommended to use a different video in the background of the large gaze contingent simulation and looking into mouse tracking as an option when the simulation gets to jumpy.
Location project: Koninklijke Visio
External supervisor: Jan Koopman
UU Supervisor: Krista Overvliet

Name Student: Frits van Iddekinge
Module: Thesis
Title project: Driving performance of elderly people with Alzheimer’s disease.
Abstract: For my thesis at SWOV I worked on an already existing project called FitCI (fitness to drive with cognitive impairments). This project was about predicting the fitness to drive of elderly people with Alzheimer’s disease (AD). It used to be the case that people with AD were not allowed to drive, but after 2009 regulations changed and people with AD were allowed to drive if they passed an on-road assessment of the CBR. The purpose of the project was to measure whether people with AD were fit to drive, without the on-road assessment. To do this, several tests (e.g. reaction time tests and drives in a driving simulator) were developed. For my thesis I focused on the difference in performance on the driving simulator between healthy people and people with AD.
Location project: SWOV
External supervisor: Michelle Doumen
UU Supervisor: Stefan van der Stigchel

<u>Poster Presentations</u>

Name Student: Floor Elbertse
Module: Internship
Title project: Visualising taste: Exploring factors in crossmodal correspondence between taste and shape
Abstract: Recent research has shown that not only do people associate sweet foods with round shapes and other tastes with more angular shapes, there are other attributes which influence this crossmodal correspondence. In an online experiment, these attributes (roundness, symmetry and number of elements) are measured in a shape-taste matching task, using food stimuli of different tastes and intensities. No significant effect has been found in this study in the relation between sweetness and roundness and symmetry or the number of elements. There did not emerge a pattern from the measured reaction times. These results are not in line with most research, which may be attributed to the choice of food stimuli or the method of instructing the participants.
Location project: UU/TNO
External supervisor: Lex Toet, TNO Soesterberg
UU Supervisor: Stella Donker

Name Student: Roderic Hillege
Title project: Watching trees – Classifying cognitive tasks from eye movements using machine learning
Module: Internship
Abstract: Is it possible to detect the cognitive task being performed solely based on the eye movement data? Humans can do it, and on a clinical population it is done successfully as well. In this study the possibility of automating classification of cognitive states using machine learning is evaluated. A dataset containing the eye movement from a natural viewing task watching television was used. A random forest machine learning model was built to classify the data. The results show that with a mean accuracy of 76.1%, eye movements could be correctly classified to the cognitive task they belong to. This promises a potent new tool for automating the analysis of cognitive eye-tracking research.
Location project: UU
External supervisor: N.a.
UU Supervisor: Stefan van der Stigchel

Name Student: Thomas Camminga
Module: Internship
Title project: Testing and designing medical e-learning courses

Abstract:From September 2018 until January 2019 I did an internship with Prolira. Prolira is developing an algorithm-based medical instrument, called the DeltaScan, that will calculate a delirium chance (i.e., acute confusional state that is very prevalent in hospitalized patients) based on EEG measurements. My goal was to assist in the development of e-learning courses about delirium and the DeltaScan, that will be nurses will complete before they start working with the device. I have conducted three usability tests throughout the process. I tested the exercises and feedback, the visual aspects and the full demo of the e-learning course on delirium. I have also developed a storyboard for the DeltaScan course, which I based on interviews and scientific literature in the fields of educational psychology and e-learning usability.
Location project: Prolira, Utrecht
External supervisor: -
UU Supervisor: Ignace Hooge

Name Student: Margreet Wijmans
Module: Internship
Title project: Extreme Faces; an eyetracking study
Abstract: In most eyetracking research that focuses on face perception, images of faces that have been standardised are used. Specific scan patterns have been found through averaging the scan patterns of all participants. However, on an individual level the scan patterns vary greatly, and the average scan patterns cannot be reproduced by individuals. Next to this, research with images of abnormal or unusual faces or facial characteristics has to my knowledge not been done before. This research focuses on how people perceive these unusual faces and whether they look at these faces differently as opposed to normal images. Global classification has been done for all images, using ugly/non ugly and pretty/non pretty categorizations.
Location project: Utrecht University (and Betweter festival at Tivoli)
External supervisor: Tjeerd de Boorder
UU Supervisor: Martijn Mulder

Name Student: James A. Trout.
Module: Internship
Title project: Creating a Guide for Designing Educational Materials in VR for Covince
Abstract: My project was to create a guide for users of the Covince system on how to create effective lessons in a VR environment. I chose to create this guide for VR as this was a document they didn't have yet, nor could I find a similar guide online despite my research. My methods involved doing a lot of solo research, looking at both scholarly articles in psychology, education, and technology, as well as talks from developers who design for VR systems such as Oculus and/or PSVR. From there, I broke down all possible educational lessons into four kinds: Task Simulators, Social Simulators, Virtual Classrooms, and Educational Environments. I further broke these down into other categories (such as how to

handle feedback and animation) and turned these long combinations of different research topics into easily-understood chunks. For my final deliverable, I added all of these into a "roadmap" in powerpoint and all my sources, linked those sources to a document with summaries of all my research, and sent in both documents. It will be posted to their website at some point and made public.
Location project: Covince.com
External supervisor: Richard Van Tilborg.
UU Supervisor: Estrella R. Montoya.

Name Student: Kazuya Takahashi.
Module: Internship
Title project: Does VR enhance listening skills?
Abstract: The rapid advances in and popularity of ICT tools has led many researchers in academia as well as teachers at school to explore their application in education across disciplines. In the early stage, VR generally targeted only limited content areas or domains of learning, but due to its full integration of artificial intelligence and a variety of social communication tools, researchers have conducted various kinds of studies to explore its potential of the dynamic learning environments. However, most of the research literature focuses on meta-analysis, and there are a few actual experiments of using VR at school settings. In addition, they criticized VR as “high-spec” products for schools.

The experiment at a Japanese school shows that there is no significant difference in scores between the VR group and the CD group. Many researchers also have long criticized VR because VR is a highly-demanding product and requires new tech literacy. However, the result of this experiment indicates that the downside of VR is not that simple. Rather, it is way advanced from the reality to which students are used to. VR is too immersive for the students to concentrate on their task. Also, VR does not prepare the students for the actual exams they would have to take in the future.

The experiment also shows that VR has great potentials for learning. Many students described VR as “imaginative” or “futuristic,” providing a positive attitude toward the use of VR in education. Thus, it is adults’ responsibility to give students more opportunities to explore the potential of VR and ignite their curiosity to challenge what education in the future means. We all have realized that great advanced in VR technology would open new horizons in the field of education for both teachers and students.
Location project: Covince.com
External supervisor: Richard Van Tilborg.
UU Supervisor: Estrella R. Montoya.

Name Student: Stefan Gaillard
Module: Internship
Title project: Experiencing is active perceiving: the influence of immersive technologies on the experience of autonomous sensory meridian response

Abstract: There is very little scientific knowledge about ASMR experiences, yet developments with regards to ASMR-inducing videos continue. Recent increases in Virtual Reality technology have enabled the creation of ASMR videos within a VR environment. If one takes into account current literature on the relationship between immersion and intensity, and reported increased immersion in VR environments, it seems plausible that ASMR experiences are more intense in a VR environment as well. This research focused on investigating the experienced intensity in VR with 3D videos and found no significant difference compared to watching videos on a screen. Several fruitful lines of new inquiry were discovered over the course of the experiment.
Location project: Utrecht University
External supervisor: -
UU Supervisor: Anouk Keizer